Aromaticity Summary

1. Aromaticity Basics

Aromatic compounds have benzene-like stability and structure.

2. Structure & Stability of Benzene

- Cyclic, planar, delocalized π electrons
- Very stable due to resonance
- Undergo substitution reactions

3. Aromaticity Conditions

- Cyclic
- Planar
- Fully conjugated
- Follows Huckel’s Rule ($4n + 2$ π electrons)

4. Classification

- Aromatic
- Non-aromatic
- Anti-aromatic

5. Key Examples

- Cyclopropenyl cation: aromatic
- Cyclobutadiene: anti-aromatic
- Cyclopentadienyl anion: aromatic
- Tropylium: aromatic
- Cyclooctatetraene: non-aromatic

6. Resonance & Stability

Aromatic > Non-aromatic > Anti-aromatic

7. Heterocyclic Compounds

Examples: Pyridine, Pyrrole

8. Nomenclature

Ortho (1,2), Meta (1,3), Para (1,4)

9. Electrophilic Substitution

- Electrophile formation
- Attack (slow step)
- Deprotonation

10. Important Reactions

- Halogenation
- Nitration
- Sulfonation
- Friedel-Crafts reactions

11. Reactivity

Benzene is less reactive than alkenes

12. Activating/Deactivating Groups

- Activating: donate electrons (o/p)
- Deactivating: withdraw electrons (meta)

13. Orientation Rules

- Donors → ortho/para
- Withdrawers → meta

14. Inductive/Resonance Effects

+I, -I, +M, -M

15. Aryl Halides

Less reactive due to strong C-X bond

16. Nucleophilic Substitution

Requires strong EWG

17. Phenols

- Acidic
- Stabilized by resonance

18. Phenol Acidity

- Increased by EWG
- Decreased by EDG

19. Phenol Reactions

- Salt formation
- Esterification
- Electrophilic substitution

20. Aromatic Acids

- Contain COOH
- Acidity affected by substituents

21. Nitro Compounds

Prepared by nitration

22. Aromatic Amines

Less basic than aliphatic amines

23. Aldehydes & Ketones

- Aldehydes: more reactive
- Ketones: less reactive

24. Carbonyl Reactions

- Nucleophilic addition
- Reduction
- Oxidation

FINAL KEY POINTS

- Aromaticity rules
- Resonance stability
- Electrophilic substitution
- Directing effects
- Phenol acidity

- Carbonyl reactivity